

Reachable Set Based Receding Horizon Control for Dynamic Drone Interception

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Abstract—This work presents a real-time motion planning framework for intercepting a dynamically maneuvering aerial target using a bounded-kinematics model. The approach combines reachable set estimation of the target with a receding-horizon control strategy for the interceptor. Instead of relying on deterministic prediction, the target’s future motion is approximated via sampled rollouts under bounded acceleration constraints. The interceptor evaluates a discrete control library and selects actions that minimize a cost function incorporating expected separation distance, terminal error, control effort, and interception timing. The method is implemented in a 3D simulation environment with PyBullet visualization. Results demonstrate successful interception within 11 seconds under stochastic target motion, highlighting the effectiveness of sampling-based prediction in dynamic pursuit scenarios.

I. INTRODUCTION

Intercepting a maneuvering aerial target is a dynamic motion planning problem requiring prediction of the target’s future states and real time computation of pursuit trajectories. The target evolves unpredictably, making open loop trajectory planning insufficient. The interceptor must repeatedly replan as new information about the target becomes available.

The core challenge is that the target’s future trajectory is unknown, meaning the interceptor observes only the target’s current position and velocity, and must reason about the set of states the target could reach given its physical acceleration limits.

We address this problem using a combination of three techniques: (1) bounded kinematics modeling of both drones as rigid spheres with velocity and acceleration constraints, (2) Monte Carlo approximation of the target’s reachable set to represent motion uncertainty, and (3) receding horizon control with a discrete action library and multi objective cost function. The key idea is to replace a single predicted trajectory with a distribution of plausible future states, allowing the interceptor to select actions that are robust.

II. RELATED WORK

The drone interception problem sits at the intersection of pursuit–evasion games, reachable set analysis, and receding horizon motion planning.

A. UAV Interception

Several recent works address the problem of intercepting non cooperative UAVs. Moreira et al. [2] tackle interception of non cooperative UAVs using physical hardware, estimating

the adversarial drone’s position with onboard sensors and a Kalman filter. Our project simplifies the sensing problem by providing the interceptor with the target’s exact state at each time step, allowing us to focus on the planning and control aspects. Pliska et al. [3] propose two planning methods for mid air interception: a model predictive control (MPC) planner and a Fast Response Proportional Navigation (FRPN) planner, benchmarked against Pure Pursuit and standard Proportional Navigation baselines. They model both drones as point masses with bounded acceleration, similar to our use of spherical rigid bodies with kinematic constraints, and demonstrate successful interception of targets at up to 5 m/s. Their comparison of multiple guidance laws motivates our choice of a receding horizon approach over simpler reactive strategies.

B. Pursuit–Evasion and Multi Agent Interception

Eklund et al. [1] use MPC with a receding horizon planner and optimization based cost to maintain a favorable position relative to an adversarial drone in pursuit/evasion scenarios. Our framework is similar in structure, but our objective is to close to within a capture radius rather than maintain a relative positioning region. Saranovic et al. [5] address the problem of using a defensive drone swarm to intercept attacking drones entering a protected area. Their attacking drones follow automated flight paths without cooperating with defenders, analogous to our target following randomized trajectories, and interception is defined by physical proximity. A key difference is that their defenders have only partial observability, whereas our interceptor has full state knowledge.

C. Reachable Set Methods

Chen et al. [4] apply Hamilton-Jacobi reachability analysis to persistent surveillance and interception, computing reachable sets via level set methods to obtain optimal control policies. Although their approach provides formal guarantees, it requires solving PDEs that scale poorly to high-dimensional systems. Our work adopts a sampling based alternative: Monte Carlo rollouts approximate the reachable set without solving the Hamilton-Jacobi PDE, retaining the core idea of reasoning about reachable states while maintaining real time computational tractability.

D. Receding Horizon Control and Sampling Based Planning

Model predictive control and receding horizon planning are well established for systems requiring repeated replan-

ning under changing conditions [6]. Bertsekas [7] provides foundational treatment of dynamic programming and optimal control, underpinning the finite horizon optimization structure used in our planner. Sampling-based planners such as RRT [9] and RRT* [10] offer probabilistic completeness guarantees for motion planning [8]. While our method does not use tree based exploration for the interceptor, the Monte Carlo sampling of the target's reachable set shares the philosophy of approximating continuous spaces through discrete samples.

III. SYSTEM MODEL

A. State and Control Spaces

Both drones operate in a bounded three dimensional workspace $\mathcal{W} \subset \mathbb{R}^3$. Each drone is modeled as a rigid sphere with radius $r = 0.35$ m, with no aerodynamic or actuator dynamics. The state of each drone at discrete time step t is defined as:

$$x_t = \begin{bmatrix} p_t \\ v_t \end{bmatrix} \in \mathcal{X} \subset \mathbb{R}^6 \quad (1)$$

where $p_t \in \mathbb{R}^3$ is the position and $v_t \in \mathbb{R}^3$ is the velocity. The control input is the commanded acceleration $u_t \in \mathcal{U} \subset \mathbb{R}^3$.

B. Discrete Time Dynamics

Both drones are modeled using discrete time bounded kinematics:

$$x_{t+1} = x_t + v_t \Delta t \quad (2)$$

$$v_{t+1} = \text{clip}(v_t + u_t \Delta t) \quad (3)$$

where velocity and acceleration are bounded by maximum limits. The simulation operates in a 3D bounded environment with collision constraints.

C. Interception Condition

Successful interception (capture) occurs when the center to center distance between the two drones is less than or equal to the sum of their radii:

$$\|p_t^I - p_t^T\|_2 \leq r_I + r_T \quad (4)$$

IV. REACHABLE SET APPROXIMATION

A. Analytical Reachable Set

Given the target's current state (p_t^T, v_t^T) and bounded acceleration $a_{T,\max}$, the predicted center trajectory under constant velocity is:

$$p_c(k) = p_t^T + v_t^T \cdot k \Delta t \quad (5)$$

for k steps into the future. The reachable set at step k is the ball:

$$\mathcal{R}_T(k) = \{p : \|p - p_c(k)\|_2 \leq r_T(k)\} \quad (6)$$

where the reachable radius grows quadratically:

$$r_T(k) = \frac{1}{2} a_{T,\max} (k \Delta t)^2 \quad (7)$$

This analytical bound provides a worst case envelope but does not capture the distribution of likely target states within the reachable set. In practice, the target's centering bias and velocity limits make many states near the boundary of the analytical set unlikely.

B. Monte Carlo Sampling

To obtain a distributional approximation, we sample N_s target rollouts. For each sample $i \in \{1, \dots, N_s\}$ and each horizon step k :

- 1) Draw a random direction vector with components uniform in $[-1, 1]$ for x, y and $[-0.6, 0.6]$ for z (reduced vertical variance), then normalize.
- 2) Scale by a random magnitude drawn uniformly from $[0.2 a_{T,\max}, a_{T,\max}]$.
- 3) Add a centering bias: a vector pointing from the current sampled position toward the environment center, scaled by a fraction β of $a_{T,\max}$ (with $\beta = 0.5$ in our experiments).
- 4) Propagate the state forward using the clipped dynamics (2)–(3).

This produces a set of N_s sampled future trajectories:

$$\hat{\mathcal{R}}_T = \left\{ \tau^{(i)} = (x_1^{(i)}, x_2^{(i)}, \dots, x_H^{(i)}) \right\}_{i=1}^{N_s} \quad (8)$$

where H is the planning horizon in steps. The centering bias prevents sampled trajectories from clustering at workspace boundaries and reflects the assumption that the target will generally remain in the operational region. The sampled set approximates the interior distribution of the reachable set, allowing the interceptor to optimize against expected target behavior rather than only the worst case.

V. CONTROL STRATEGY

A. Receding Horizon Framework

The interceptor does not plan a single trajectory and follow it to completion. Instead, at every time step it picks the best action it can for the near future, executes one step, then replans using the latest target information. This loop repeats for the entire simulation.

B. Discrete Action Library

Rather than searching over all possible accelerations, the planner considers a fixed set of 27 candidate actions. These are generated by combining three evenly spaced values along each axis $(-1, 0, 1)$, normalizing each nonzero combination to a fixed magnitude a_{cmd} , and including a coasting (zero acceleration) option. This set is denoted $\mathcal{A} = \{a_1, a_2, \dots, a_M\}$.

C. Forward Simulation and Evaluation

For each candidate action $a_j \in \mathcal{A}$, the planner simulates the interceptor forward over a planning horizon of H steps, applying the same action at every step. The resulting trajectory is then compared against all N_s sampled target rollouts to compute a cost score.

D. Cost Function

The cost for each candidate action is:

$$J(a_j) = d_{\min} + \lambda d_{\text{term}} + \gamma \|a_j\| + \omega W(a_j) - \frac{\eta}{t_{\text{hit}} + \epsilon} \quad (9)$$

where:

- $d_{\min}$ is the minimum (over horizon steps) of the mean separation distance between the interceptor and the sampled target positions,
- d_{term} is the mean separation at the terminal time step,
- $\|a_j\|$ is the control effort (acceleration magnitude),
- $W(a_j)$ is a wall avoidance penalty: for each predicted interceptor state, if the distance to the nearest workspace boundary is less than a margin m , a penalty $(m - d_{\text{wall}})/m$ is accumulated,
- t_{hit} is the earliest time step at which the interception distance threshold is met across any sample, and ϵ is a small constant for numerical safety,
- $\lambda, \gamma, \omega, \eta$ are tunable weights.

The interceptor selects the action that minimizes this cost:

$$a^* = \arg \min_{a_j \in \mathcal{A}} J(a_j) \quad (10)$$

E. Planning Algorithm

The complete planning loop is summarized in Algorithm 1.

Algorithm 1 Receding Horizon Interception Planner

Require: Current states x_t^I, x_t^T ; horizon H ; samples N_s ; action library $\mathcal{A}$

- 1: Sample N_s target rollouts $\hat{\mathcal{R}}_T$ from current target state
- 2: **for** each action $a_j \in \mathcal{A}$ **do**
- 3: Simulate interceptor forward H steps under constant a_j
- 4: **for** each horizon step $k = 1, \dots, H$ **do**
- 5: Compute distances to all N_s target samples at step k
- 6: Track minimum mean gap, terminal gap, earliest hit
- 7: Accumulate wall avoidance penalty if near boundary
- 8: **end for**
- 9: Compute $J(a_j)$ from Eq. (11)
- 10: **end for**
- 11: Select $a^* = \arg \min_{a_j} J(a_j)$
- 12: Apply a^* for one simulation time step

VI. SIMULATION SETUP

A. Parameters

The simulation uses the parameters listed in Table I.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Simulation time step Δt	0.1 s
Horizon time step Δt_H	0.05 s
Planning horizon	2.0 s (40 steps)
Target rollout samples N_s	25
Action grid per axis N_{axis}	3
Interceptor max accel/speed	100 / 100
Target max accel/speed	20 / 20
Seeker command magnitude a_{cmd}	8.0 m/s ²
Terminal distance weight λ	0.35
Control effort weight γ	0.005
Wall avoidance weight ω	0.5
Wall margin m	2.0 m
Early intercept bonus η	8.0
Centering bias β	0.5
Drone radius (both)	0.35 m
Workspace bounds	$[-50, 50]^2 \times [1, 50]$ m
Max simulation time	35.0 s

B. Target Behavior

The target drone follows a stochastic trajectory: at each time step, a random acceleration vector is drawn from within its bounded acceleration limit (magnitude uniformly between 25% and 100% of $a_{T,\text{max}}$) with reduced vertical variance. A centering bias of $0.2 a_{T,\text{max}}$ pulls the target toward the workspace center, preventing persistent drift toward boundaries. The target's state is provided directly to the interceptor at each step (perfect observation, no sensing noise).

C. Initial Conditions

The target starts at a uniformly random position within the workspace with a random initial velocity (each component uniform in $[-5, 5]$ m/s). The interceptor starts at a fixed position $[7.5, 7.5, 5.0]$ m with zero initial velocity.

D. Visualization

PyBullet is used purely for 3D visualization of the drone trajectories. All dynamics propagation, reachable set sampling, and control computation are performed analytically in the planning code—PyBullet does not participate in the physics simulation.

E. Evaluation Protocol

The planner is evaluated over 500 independent Monte Carlo trials, each with randomized initial conditions for the target. Each trial runs until interception occurs or the maximum time limit (35 s) is reached. The following metrics are recorded: success rate, mean time to interception, mean number of simulation steps, and mean final center to center distance at capture.

VII. RESULTS

Over 500 Monte Carlo trials with stochastic target motion, the system achieves the performance summarized in Table II.

TABLE II
INTERCEPTION PERFORMANCE (500 TRIALS)

Metric	Value
Success rate	85.8%
Mean interception time	10.74 s
Mean simulation steps	107.37
Mean final distance	0.5366 m

Figs. 1–3 show distance, speed, and acceleration over time for three representative trials with different initial conditions. In all three cases, the separation distance drops steadily to zero, confirming successful interception. The speed plots show the seeker accelerating to close the gap while the target moves unpredictably. The acceleration plots show the seeker’s control input responding to changes in the target’s motion.

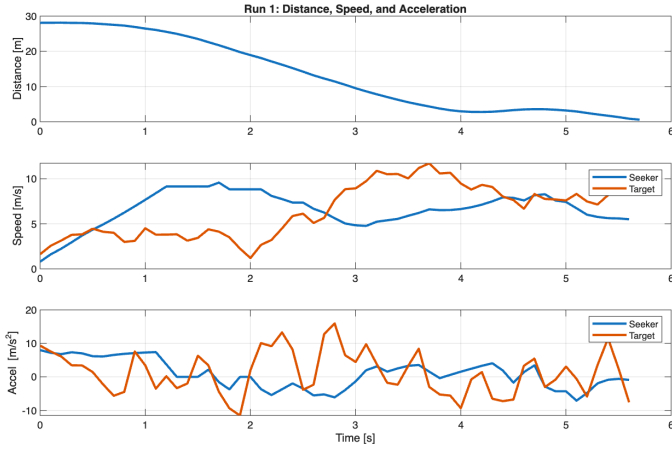


Fig. 1. Run 1: Distance, speed, and acceleration over time. The seeker intercepts the target in approximately 6 seconds from an initial separation of 30 m.

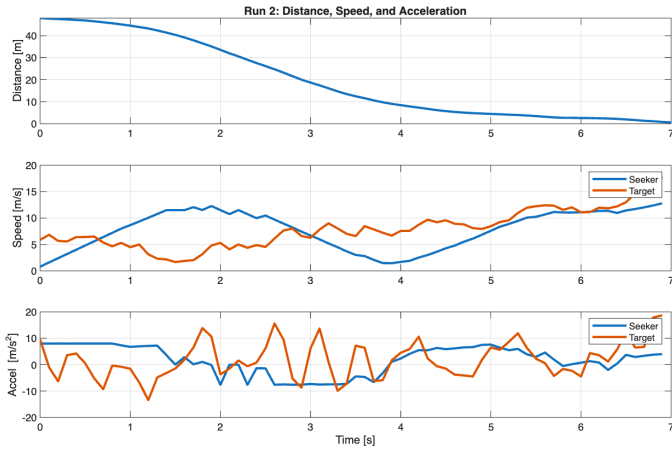


Fig. 2. Run 2: Distance, speed, and acceleration over time. The seeker intercepts the target in approximately 7 seconds from an initial separation of 40 m.

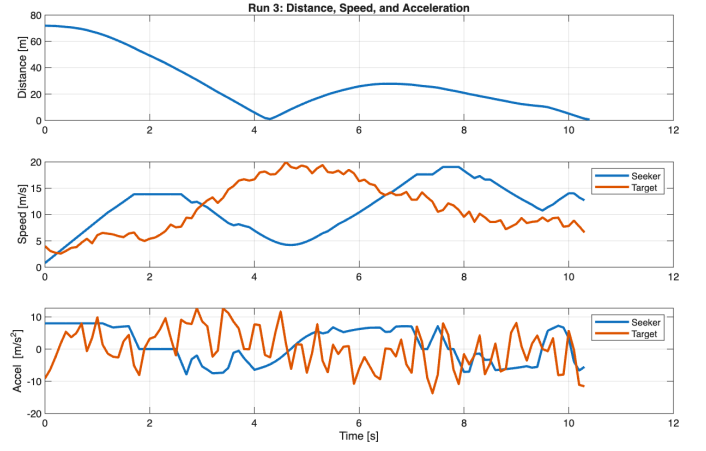


Fig. 3. Run 3: Distance, speed, and acceleration over time. The seeker intercepts the target in approximately 12 seconds from an initial separation of 60 m.

The 14.2% failure rate is attributable to cases where the target’s stochastic motion produces sharp reversals near workspace boundaries that temporarily increase separation beyond the planner’s horizon, causing the interceptor to lose its convergence advantage before the 35 second time limit.

VIII. ANALYSIS

A. Completeness

The planner is not complete in the formal sense: it does not guarantee that a solution will be found for every feasible interception scenario. The discrete action library limits the interceptor to a finite set of maneuvers at each step, and the finite sampling of the target’s reachable set may miss extreme trajectories. However, the 5:1 kinematic advantage ($a_{I,\max}/a_{T,\max}$) ensures that interception is feasible in the vast majority of configurations, and the 85.8% empirical success rate supports this.

B. Optimality

The method is not asymptotically optimal. The discrete action library and finite horizon evaluation preclude convergence to a globally optimal interception trajectory. The cost function balances multiple objectives, so the resulting trajectory minimizes a surrogate cost rather than a theoretically optimal criterion such as minimum time interception. Notably, the seeker command magnitude ($a_{\text{cmd}} = 8.0 \text{ m/s}^2$) is well below the seeker’s physical limit ($a_{I,\max} = 100 \text{ m/s}^2$), meaning the planner intentionally under actuates the interceptor for smoother pursuit at the cost of time optimality.

C. Computational Efficiency

At each time step, the planner evaluates $|\mathcal{A}| \times N_s$ forward simulations, each of length $H = 40$ steps. With $|\mathcal{A}| = 27$ candidate actions and $N_s = 25$ samples, this yields approximately 27,000 state propagations per planning cycle. Each propagation involves only linear arithmetic (vector addition, norm computation, clipping), making the computation real time feasible at the 0.1 s simulation time step.

D. Robustness

The primary strength of the approach is its robustness to stochastic target motion. By evaluating candidate actions against a sampled distribution of target futures rather than a single predicted trajectory, the planner hedges against uncertainty. The centering bias in the target sampling reflects a soft prior about target behavior, concentrating samples in the most likely region of the reachable set rather than uniformly covering the worst case boundary.

E. Comparison to Alternative Approaches

A purely proportional navigation approach—steering toward the target’s current position, as used as a baseline in [3]—would lack predictive capability and suffer from lag in response to target maneuvers. A Hamilton–Jacobi reachability approach [4] would provide formal guarantees but is computationally prohibitive in real time for a 6D state space. The MPC approach of [1] optimizes over a continuous action space but assumes a more structured adversary model. Our sampling based method occupies a middle ground: more robust than reactive pursuit, more tractable than exact reachability computation, and simpler to implement than full continuous MPC.

IX. CONCLUSION

This work demonstrates that combining Monte Carlo reachable set sampling with receding horizon control provides an effective framework for dynamic drone interception under uncertainty. The method achieves an 85.8% success rate over 500 stochastic trials while maintaining real time computational feasibility, validating the core hypothesis that distributional prediction of target motion is superior to deterministic point prediction for pursuit planning.

The key lessons from this project are threefold. First, replacing a single trajectory prediction with a sampled reachable set significantly improves robustness to unpredictable target motion. Second, a simple discrete action library combined with a well designed multi objective cost function can produce effective interception behavior without continuous optimization. Third, the receding horizon structure provides the necessary feedback to correct for prediction errors as new target state information becomes available.

Several directions for future work are promising:

- **RRT* based global planning:** Integrating a sampling based global planner (e.g., RRT*) with the local receding horizon controller could improve performance in environments with obstacles or when longer horizon reasoning is needed.
- **Adaptive sampling:** Dynamically adjusting the number of target rollout samples based on the current uncertainty level could improve both efficiency and robustness.
- **Learning based prediction:** Replacing the uniform random rollouts with a learned model of target behavior could yield tighter reachable set estimates and faster convergence.

- **Multi interceptor coordination:** Extending the framework to multiple interceptor drones cooperating to capture one or more targets is a natural next step for practical counter UAS applications.

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